

Series — Question Bank

Engineering Mathematics–I (CSE-1001)

BTech 1st Year, IIIT Bhopal

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Instructions

This bank accompanies the note *Series*. Before attempting it, make sure you have worked through *Sequences* and its question bank.

How the parts are meant to be used:

1. **Part A (MCQ)** — rapid revision. One minute per question.
2. **Part B (True or False)** — every answer must carry a reason or a counterexample.
3. **Part C (Short answer)** — the routine tests, one per question, exactly as they appear in an examination.
4. **Part D (Long answer)** — statements of theorems, proofs, and the longer multi-case problems, including series containing a parameter x .
5. **Part E (Critical thinking)** — traps, counterexamples and borderline cases.

One habit worth forming. Before applying any test, run down the strategy list: does $a_n \rightarrow 0$? is it geometric or a p -series? does it telescope? are there factorials or n -th powers? is it a rational function of n ? does the sign alternate? Choosing the right test is most of the work, and in an examination it is also most of the time saved.

Answers and full worked solutions are issued as a separate document, *Series* — *Answer Key*.

Part A: Multiple Choice Questions

46 questions $\times$ 1 mark

Choose the one correct option.

Q1. The statement “ $\sum a_n$ converges” means that the sequence of

- (a) terms a_n converges
- (b) partial sums $s_n = a_1 + \cdots + a_n$ converges
- (c) ratios a_{n+1}/a_n converges
- (d) absolute values $|a_n|$ converges

Q2. For $\sum_{n=1}^{\infty} \frac{1}{2^n}$, the n -th partial sum is

- (a) $1 - \frac{1}{2^n}$
- (b) $2 - \frac{1}{2^n}$
- (c) $\frac{1}{2^n}$
- (d) $\frac{n}{2}$

Q3. The geometric series $\sum_{n=0}^{\infty} ar^n$ ($a \neq 0$) converges if and only if

- (a) $|r| \leq 1$
- (b) $|r| < 1$
- (c) $r > 0$
- (d) $|r| \geq 1$

Q4. For $|r| < 1$, $\sum_{n=0}^{\infty} ar^n$ equals

- (a) $\frac{a}{1+r}$
- (b) $\frac{a}{1-r}$
- (c) $\frac{ar}{1-r}$
- (d) $\frac{1}{1-r}$

Q5. $\frac{1}{9} + \frac{1}{27} + \frac{1}{81} + \cdots$ equals

- (a) $\frac{1}{6}$
- (b) $\frac{1}{3}$
- (c) $\frac{1}{9}$
- (d) $\frac{1}{2}$

Q6. The repeating decimal $0.3333\ldots$ equals

- (a) $\frac{1}{3}$
- (b) $\frac{3}{10}$
- (c) $\frac{1}{30}$
- (d) 3

Q7. $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ equals

- (a) 0
- (b) 1
- (c) $\frac{1}{2}$
- (d) it diverges

Q8. If $\lim_{n \rightarrow \infty} a_n \neq 0$, then $\sum a_n$

- (a) converges
- (b) diverges
- (c) may converge or diverge
- (d) converges absolutely

Q9. If $\lim_{n \rightarrow \infty} a_n = 0$, then $\sum a_n$

- (a) converges
- (b) diverges
- (c) may converge or may diverge
- (d) converges conditionally

Q10. The harmonic series $\sum \frac{1}{n}$

- (a) converges to 1
- (b) converges to $\ln 2$
- (c) diverges
- (d) oscillates

Q11. By the p -test, $\sum \frac{1}{n^p}$ converges if and only if

- (a) $p < 1$
- (b) $p \leq 1$
- (c) $p > 1$
- (d) $p \geq 1$

Q12. The series $\sum \frac{1}{\sqrt{n}}$

- (a) converges
- (b) diverges
- (c) converges conditionally
- (d) oscillates

Q13. $\sum_{n=1}^{\infty} \frac{1}{n^2}$ equals

- (a) $\frac{\pi^2}{6}$
- (b) $\frac{\pi}{4}$
- (c) e
- (d) 1

Q14. The Leibniz series $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots$ equals

- (a) $\frac{\pi}{4}$
- (b) $\frac{\pi^2}{6}$
- (c) $\ln 2$
- (d) 1

Q15. $\sum_{n=1}^{\infty} \frac{1}{n!}$ equals

- (a) e
- (b) $e - 1$
- (c) 1
- (d) $\frac{\pi^2}{6}$

Q16. The comparison test may be applied only when the terms are

- (a) alternating
- (b) non-negative
- (c) decreasing
- (d) bounded

Q17. For $\sum \frac{1}{n^2 + 1}$, comparison with $\frac{1}{n^2}$ shows the series is

- (a) convergent
- (b) divergent
- (c) inconclusive
- (d) oscillatory

Q18. For $\sum \frac{1}{\sqrt{n^3 + 1}}$, the natural comparison series is

- (a) $\sum \frac{1}{n}$
- (b) $\sum \frac{1}{n^{3/2}}$
- (c) $\sum \frac{1}{n^2}$
- (d) $\sum \frac{1}{n^3}$

Q19. If $a_n, b_n > 0$ and $\lim \frac{a_n}{b_n} = c$ with $0 < c < \infty$, then

- (a) both series converge
- (b) both series diverge
- (c) both series converge or both diverge
- (d) no conclusion is possible

Q20. For $\sum \frac{2n-1}{n(n+1)(n+2)}$, the natural choice of v_n in the limit comparison test is

- (a) $\frac{1}{n}$
- (b) $\frac{1}{n^2}$
- (c) $\frac{1}{n^3}$
- (d) $\frac{1}{\sqrt{n}}$

Q21. In D'Alembert's ratio test, $\rho < 1$ implies the series is

- (a) divergent
- (b) convergent
- (c) oscillatory
- (d) inconclusive

Q22. In D'Alembert's ratio test, $\rho = 1$ means

- (a) the series converges
- (b) the series diverges
- (c) the test fails and another method is needed
- (d) the series converges absolutely

Q23. For $\sum \frac{n!}{n^n}$, the ratio ρ equals

- (a) 1
- (b) e
- (c) $\frac{1}{e}$
- (d) 0

Q24. For $\sum \frac{n!}{10^n}$, the ratio ρ equals

- (a) 0
- (b) $\frac{1}{10}$
- (c) ∞
- (d) 1

Q25. Cauchy's root test examines the limit of

- (a) $\frac{a_{n+1}}{a_n}$
- (b) $(a_n)^{1/n}$
- (c) $\int f$
- (d) $|a_n|$

Q26. For $\sum \frac{n^3}{3^n}$, the root test gives ρ equal to

- (a) $\frac{1}{3}$
- (b) 3
- (c) 1
- (d) 0

Q27. For $\sum \left(\frac{n+1}{n+2}\right)^n x^n$ with $x > 0$, the root test gives ρ equal to

- (a) 1
- (b) x
- (c) $\frac{x}{e}$
- (d) $\frac{1}{e}$

Q28. The integral test requires f to be positive, continuous and

- (a) increasing
- (b) decreasing
- (c) bounded
- (d) alternating

Q29. The series $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$

- (a) converges
- (b) diverges
- (c) converges conditionally
- (d) cannot be tested

Q30. By Leibniz's test, $\sum (-1)^{n+1} u_n$ with $u_n > 0$ converges if (u_n) is non-increasing and

- (a) $u_n \rightarrow 1$
- (b) $u_n \rightarrow 0$
- (c) $\sum u_n$ converges
- (d) u_n is bounded

Q31. The alternating harmonic series $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$ converges to

- (a) 0
- (b) $\ln 2$
- (c) $\frac{\pi}{4}$
- (d) 1

Q32. The series $\sum \frac{(-1)^{n+1}}{n}$ is

- (a) absolutely convergent
- (b) conditionally convergent
- (c) divergent
- (d) oscillatory

Q33. The series $\sum \frac{(-1)^{n+1}}{n^2}$ is

- (a) absolutely convergent
- (b) conditionally convergent
- (c) divergent
- (d) oscillatory

Q34. If $\sum |a_n|$ converges, then $\sum a_n$

- (a) diverges
- (b) converges
- (c) gives no information
- (d) is conditionally convergent

Q35. If $\sum |a_n|$ diverges, then about $\sum a_n$ we may conclude

- (a) it diverges
- (b) it converges
- (c) nothing without further work
- (d) it is absolutely convergent

Q36. In Leibniz's test, the error after n terms satisfies $|S - s_n| \leq$

- (a) u_n
- (b) u_{n+1}
- (c) s_n
- (d) $\frac{1}{n}$

Q37. The series $\sum_{n=1}^{\infty} \frac{\sin nx}{n^2}$ is

- (a) divergent for all x
- (b) absolutely convergent for all x
- (c) conditionally convergent only
- (d) oscillatory

Q38. The series $\sum \frac{(-1)^{n-1}}{2n-1}$ is

- (a) absolutely convergent
- (b) conditionally convergent
- (c) divergent
- (d) equal to e

Q39. Which of these series converges?

- (a) $\sum \frac{n}{1+n}$ (b) $\sum (-1)^{n+1}$ (c) $\sum n^2$ (d) $\sum \frac{1}{n^2}$

Q40. The series $\sum_{n=0}^{\infty} \frac{2^{n+5}}{3^n}$

- (a) converges
- (b) diverges
- (c) oscillates
- (d) is conditionally convergent

Q41. The series $\sum_{n=1}^{\infty} \frac{(2n)!}{n!n!}$

- (a) converges
- (b) diverges
- (c) oscillates
- (d) cannot be decided by the ratio test

Q42. A power series $\sum c_n(x-a)^n$ with radius of convergence R converges absolutely for

- (a) all x (b) $|x-a| < R$ (c) $|x-a| > R$ (d) $x = a$ only

Q43. The interval of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n}$ is

- (a) $(-1, 1)$ (b) $[-1, 1]$ (c) $[-1, 1)$ (d) $(-1, 1]$

Q44. The Maclaurin series of e^x is

- (a) $\sum \frac{x^n}{n!}$ (b) $\sum x^n$ (c) $\sum \frac{(-1)^n x^n}{n!}$ (d) $\sum n! x^n$

Q45. The radius of convergence of $\sum_{n=0}^{\infty} \frac{(x-2)^n}{3^n}$ is

- (a) 1 (b) 2 (c) 3 (d) ∞

Q46. Zeno's sum $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots$ equals

- (a) $\frac{1}{2}$ (b) 1 (c) 2 (d) ∞

Part B: True or False — with justification10 questions $\times$ 1 mark

State whether each is true or false. **A verdict without a reason or a counterexample earns no marks.**

- Q1.** If $a_n \rightarrow 0$, then $\sum a_n$ converges.
- Q2.** If $\sum a_n$ converges, then $a_n \rightarrow 0$.
- Q3.** Every absolutely convergent series is convergent.
- Q4.** Every convergent series is absolutely convergent.
- Q5.** The series $\sum \frac{1}{n^p}$ converges when $p = 1$.
- Q6.** The comparison test may be applied directly to $\sum \frac{(-1)^n}{n}$.
- Q7.** If the ratio test gives $\rho = 1$, the series diverges.
- Q8.** A geometric series with $r = 1$ converges.
- Q9.** Rearranging the terms of a conditionally convergent series cannot change its sum.
- Q10.** If $\sum a_n$ and $\sum b_n$ both diverge, then $\sum(a_n + b_n)$ diverges.

Part C: Short Answer Questions20 questions $\times$ 3 marks

- Q1.** Define the sequence of partial sums of $\sum a_n$. For $a_n = \frac{1}{2^n}$, find s_n in closed form and hence the sum of the series.
- Q2.** Examine the convergence of $\frac{1}{9} + \frac{1}{27} + \frac{1}{81} + \cdots$, and find the sum if it converges.
- Q3.** Examine the convergence of $1 - \frac{1}{3} + \frac{1}{3^2} - \frac{1}{3^3} + \cdots$, and find the sum.
- Q4.** Evaluate $\sum_{n=0}^{\infty} \frac{(-1)^n 5}{4^n}$.
- Q5.** Show that $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \cdots$ converges, and find its sum.
- Q6.** Test $\sum_{n=1}^{\infty} \frac{n}{1+n}$ for convergence.
- Q7.** Test $\sum_{n=1}^{\infty} (-1)^{n+1}$ and $\sum_{n=1}^{\infty} \frac{-n}{2n+5}$ for convergence.

- Q8.** Using the p -test, decide the behaviour of $\sum \frac{1}{n^{3/2}}$, $\sum \frac{1}{n^{2/3}}$ and $\sum \frac{1}{n}$.
- Q9.** Test $\sum_{n=1}^{\infty} \frac{1}{n^2 + 1}$ using the comparison test.
- Q10.** Test $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^3 + 1}}$ using the comparison test.
- Q11.** Test $\frac{1}{1 \cdot 2 \cdot 3} + \frac{3}{2 \cdot 3 \cdot 4} + \frac{5}{3 \cdot 4 \cdot 5} + \dots$ using the limit form of the comparison test.
- Q12.** Test $1 + \frac{1}{2^2} + \frac{2^2}{3^3} + \frac{3^3}{4^4} + \dots$ using the limit form of the comparison test.
- Q13.** Test $\sum_{n=1}^{\infty} \frac{n!}{n^n}$ using D'Alembert's ratio test.
- Q14.** Test $\sum_{n=1}^{\infty} \frac{n!}{(n^n)^2}$ using D'Alembert's ratio test.
- Q15.** Test $\sum \frac{n^3}{3^n}$ using Cauchy's root test.
- Q16.** Test $\sum_{n=1}^{\infty} \left(\frac{n}{n+1} \right)^{n^2}$ using Cauchy's root test.
- Q17.** Check the convergence of $1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{4}} + \dots$
- Q18.** Classify $\sum \frac{(-1)^{n+1}}{n}$ and $\sum \frac{(-1)^{n+1}}{n^2}$ as absolutely convergent, conditionally convergent or divergent.
- Q19.** Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1}$ is absolutely or conditionally convergent.
- Q20.** Examine the convergence of $\sum_{n=1}^{\infty} \frac{\sin nx}{n^2}$, where $x \in \mathbb{R}$ is fixed.

Part D: Long Answer Questions

13 questions $\times$ 6 marks

- Q1.** Define the convergence of an infinite series through its partial sums. Hence prove that $\sum_{n=0}^{\infty} ar^n$ converges to $\frac{a}{1-r}$ when $|r| < 1$, and examine separately what happens when $r = 1$, $r = -1$ and $|r| > 1$.
- Q2.** Prove that the harmonic series $\sum \frac{1}{n}$ diverges, by grouping the terms in blocks of length 1, 1, 2, 4, 8, ... Contrast this with $\sum \frac{1}{n^2}$ and explain what makes the two behave differently.

Q3. State and prove the n -th term test for divergence. Show by two examples that the converse is false, and explain in one sentence why the test can never be used to prove convergence.

Q4. State the p -test. Prove it using the integral test, treating the cases $p > 1$, $p = 1$ and $p < 1$ separately. Apply the result to $\sum \frac{1}{n^{5/2}}$, $\sum \frac{1}{n}$, $\sum \frac{1}{n^{0.9}}$ and $\sum \frac{1}{\sqrt[3]{n}}$.

Q5. State the comparison test and its limit form. Use whichever is convenient to test

$$\begin{aligned} \text{(i)} \quad & \sum \frac{1}{n^2 + 1} & \text{(ii)} \quad & \frac{1}{4 \cdot 7 \cdot 10} + \frac{4}{7 \cdot 10 \cdot 13} + \frac{9}{10 \cdot 13 \cdot 16} + \cdots \\ \text{(iii)} \quad & \sum_{n=1}^{\infty} \frac{1}{\sqrt{n} + \sqrt{n+1}} & \text{(iv)} \quad & \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{1}{n} \end{aligned}$$

Q6. State D'Alembert's ratio test and use it to discuss the convergence of

$$\text{(i)} \quad \sum_{n=1}^{\infty} \frac{n! 3^n}{n^n} \quad \text{(ii)} \quad \sum_{n=1}^{\infty} \left(\frac{n^2}{2^n} + \frac{1}{n^2} \right) \quad \text{(iii)} \quad \sum_{n=1}^{\infty} \frac{n^3 - n + 1}{n!}$$

Q7. Discuss the convergence of

$$\frac{1}{2\sqrt{1}} + \frac{x^2}{3\sqrt{2}} + \frac{x^4}{4\sqrt{3}} + \frac{x^6}{5\sqrt{4}} + \cdots$$

for all values of x . Treat the boundary case separately and name the test used at each stage.

Q8. Discuss the nature of each of the following series for $x > 0$, treating the case $x = 1$ separately:

$$\text{(i)} \quad \frac{1}{2} + \frac{2}{3}x + \left(\frac{3}{4}\right)^2 x^2 + \left(\frac{4}{5}\right)^3 x^3 + \cdots \quad \text{(ii)} \quad \sum_{n=1}^{\infty} \frac{(n+1)^n x^n}{n^{n+1}}$$

Q9. State Leibniz's test in full, listing all its conditions. Apply it to

$$\begin{aligned} \text{(i)} \quad & \frac{5}{2} - \frac{7}{4} + \frac{9}{6} - \frac{11}{8} + \cdots & \text{(ii)} \quad & \sum_{n=1}^{\infty} \frac{(-1)^n}{n!} & \text{(iii)} \quad & \frac{1}{6} - \frac{2}{11} + \frac{3}{16} - \frac{4}{21} + \cdots \\ \text{(iv)} \quad & \frac{1}{1 \cdot 2} - \frac{1}{3 \cdot 4} + \frac{1}{5 \cdot 6} - \frac{1}{7 \cdot 8} + \cdots \end{aligned}$$

Q10. Define absolute and conditional convergence, and state the theorem connecting absolute convergence with convergence. Test each of the following for convergence and, where convergent, say which kind:

$$\text{(i)} \quad \sum \frac{(-1)^{n-1}}{n\sqrt{n}} \quad \text{(ii)} \quad \sum \frac{(-1)^{n-1}n^2}{2^n} \quad \text{(iii)} \quad \sum \frac{(-1)^{n-1}n}{n^2 + 1} \quad \text{(iv)} \quad \sum \frac{(-1)^n 3^n}{n!}$$

Q11. State the integral test and explain, with a sketch of rectangles against the curve $y = f(x)$, why it is true. Use it to settle $\sum \frac{1}{n^2}$ and $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$.

- Q12.** Define the radius and interval of convergence of a power series. Find both, with the endpoints tested separately, for

$$(i) \sum_{n=1}^{\infty} \frac{x^n}{n} \quad (ii) \sum_{n=0}^{\infty} \frac{(x-2)^n}{3^n} \quad (iii) \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

Write down also the Maclaurin series of e^x , $\sin x$ and $\cos x$.

- Q13.** Write down the strategy you would follow to decide the convergence of an arbitrary series, as an ordered list of checks. Then apply it, naming the test used at each step, to

$$(i) \sum \frac{1}{n!} \quad (ii) \sum \frac{2^{n+5}}{3^n} \quad (iii) \sum \frac{4^n n! n!}{(2n)!}$$

$$(iv) \sum \frac{n-2}{n^3 - n^2 + 3} \quad (v) \sum \sqrt{\frac{n+1}{n^2 + 2}}$$

Part E: Critical Thinking Questions

8 questions $\times$ 5 marks

- Q1.** A set of notes states: “The series $\sum_{n=1}^{\infty} ar^{n-1}$ is convergent if $|r| \leq 1$ and divergent if $|r| \geq 1$.” Show that this statement is self-contradictory, identify the value of $|r|$ at which it goes wrong, and write out the correct statement. Then show that convergence fails at $r = 1$ and at $r = -1$ for two *different* reasons.
- Q2.** Both $\sum \frac{1}{n}$ and $\sum \frac{1}{n^2}$ have $a_n \rightarrow 0$, and both give $\rho = 1$ in D’Alembert’s ratio test. One diverges and the other converges. Explain why no version of the ratio test could possibly decide between them, and settle each by an appropriate method.
- Q3.** The series $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$ converges to $\ln 2$, while $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$ diverges. Explain what the minus signs are contributing, state the technical name for this situation, and say what would happen to the first series if all its terms were replaced by their absolute values.
- Q4.** Explain why the phrase “the sum of the series” is ambiguous for a conditionally convergent series but unambiguous for an absolutely convergent one. Illustrate with the alternating harmonic series, whose terms can be rearranged to give $\frac{3}{2} \ln 2$ instead of $\ln 2$. What practical rule does this suggest when manipulating series?
- Q5.** Give an example of two divergent series $\sum a_n$ and $\sum b_n$ for which $\sum(a_n + b_n)$ converges. Explain carefully why this does not contradict any theorem you know, and state the correct hypothesis under which $\sum(a_n + b_n) = \sum a_n + \sum b_n$ is valid.
- Q6.** A student argues: “For $\sum \frac{1}{n}$, the terms $\frac{1}{n}$ tend to 0, and consecutive partial sums satisfy $s_{n+1} - s_n = \frac{1}{n+1} \rightarrow 0$. Hence (s_n) is a Cauchy sequence and the series converges.” The series in fact diverges. Identify *both* errors in this argument, and support the second by computing $s_{2n} - s_n$.

- Q7.** For which values of $k > 0$ does $\sum_{n=1}^{\infty} \frac{(n!)^2 k^n}{(2n)!}$ converge? Use the ratio test to find the threshold, state what happens strictly below and strictly above it, and explain why the threshold value itself needs a separate argument.
(You are not required to settle the threshold case completely; explain why the ratio test cannot settle it.)
- Q8.** A student writes: “Since $\frac{1}{n} < \frac{1}{\sqrt{n}}$ for all $n \geq 2$, and $\sum \frac{1}{\sqrt{n}}$ diverges, it follows that $\sum \frac{1}{n}$ diverges.” The conclusion happens to be correct. Is the *reasoning* valid? Explain precisely which half of the comparison test has been misused, and give a correct comparison argument for the same conclusion.

Answers: see **Series — Answer Key**